

The empirical recurrence for OEIS A231129: recovering a conjecture that is not written down

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Abstract

OEIS A231129 records “Empirical recurrence of order 90”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 191$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 90, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 90 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A231129 is “Number of $(n+1) \times (6+1)$ white-square subarrays of 0..2 arrays with no element equal to a strict majority of its diagonal and antidiagonal neighbors, with values 0..2 introduced in row major order.”. It has offset 1 and begins

120, 3584, 124644, 4150352, 137063228, 4614346288, 151928497692, 5118452297664, ...

The entry states:

Empirical recurrence of order 90 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:16:12, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4645$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 191$ of the 4645, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 191$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 9290$ exact terms of the model returns order 90 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{90} c_i a(n-i),$$

c_1	0	c_{24}	−12825439835499940267	c_{47}	0	c_{70}	43
c_2	1416	c_{25}	0	c_{48}	8343622817568076810943584	c_{71}	−12
c_3	0	c_{26}	22220478074044956658	c_{49}	0	c_{72}	−12
c_4	−380608	c_{27}	0	c_{50}	−11999700040375582762439200	c_{73}	−7
c_5	0	c_{28}	180368148353595896317	c_{51}	0	c_{74}	−7
c_6	47744341	c_{29}	0	c_{52}	19422137520412861782007040	c_{75}	16
c_7	0	c_{30}	−1763982873783235634082	c_{53}	0	c_{76}	16
c_8	−3556942510	c_{31}	0	c_{54}	−31945484195327878169154816	c_{77}	−3
c_9	0	c_{32}	8088464233360046642647	c_{55}	0	c_{78}	−3
c_{10}	175549260169	c_{33}	0	c_{56}	42419427868128265947866112	c_{79}	9
c_{11}	0	c_{34}	−23765208872019635601556	c_{57}	0	c_{80}	9
c_{12}	−6077949309941	c_{35}	0	c_{58}	−38259999884960774369391104	c_{81}	−
c_{13}	0	c_{36}	50818243282715888025227	c_{59}	0	c_{82}	−
c_{14}	151893306061265	c_{37}	0	c_{60}	17399205884214230298245120	c_{83}	3
c_{15}	0	c_{38}	−107864621335637733913594	c_{61}	0	c_{84}	3
c_{16}	−2781653897416825	c_{39}	0	c_{62}	4404538206847081838673920	c_{85}	−
c_{17}	0	c_{40}	327241023359107679535790	c_{63}	0	c_{86}	−
c_{18}	37710018762882743	c_{41}	0	c_{64}	−12885641664200647465910272	c_{87}	−
c_{19}	0	c_{42}	−1077923411555406788081432	c_{65}	0	c_{88}	−
c_{20}	−378575249660309773	c_{43}	0	c_{66}	11350657234040641423998976	c_{89}	−
c_{21}	0	c_{44}	2806584853644759842930392	c_{67}	0	c_{90}	−
c_{22}	2735765984965656744	c_{45}	0	c_{68}	−7942433027984578871164928		
c_{23}	0	c_{46}	−5446875045218451815317616	c_{69}	0		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 90, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $90 + 4$ terms removed returns order 90 as well, so $\deg q_{\min} = 90 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $90 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 90 that this sequence satisfies — and that family turns out to have exactly one member.

References

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